SINGAPORE DIMENSIONS • GRADE 3A • CHAPTER 5

Multi-Step Word Problems & Bar Models

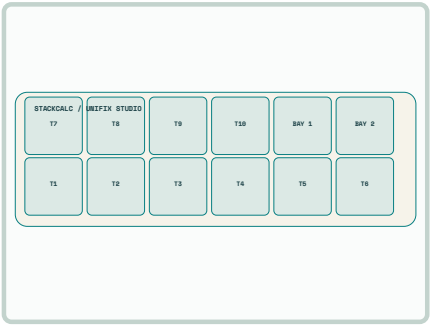
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.OA.D.8

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Build tape models with Counting & Array tiles: 24 total, subtract 6, split into 3.

Sketch Model on Paper:
Draw Singapore bar model: total bar with bracket and partition.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1. 2 **Step 1** Execute [2]
2. 4 **Step 2** Execute [4]
3. ENTER **Step 3** Execute [ENTER]
4. 6 **Step 4** Execute [6]

EXPECTED LCD X: 6 stickers each

1 **Sarah has $50, buys 2 shirts at $15 each. Change?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

2 **A baker bakes 40 loaves, sells 16, packs remainder in 4 bins.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

3 **5 boxes with 8 pencils each, plus 12 extra loose pencils.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

4 **Three friends each buy a $4 snack, and split a $9 pizza. Total cost?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer:

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 5

Multi-Step Word Problems & Bar Models • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Counting & Array Studio (COUNT-01) & 4-Line CAD Stack

CAD Blueprint: unifix_tray

Target Device: SC-6 Elementary

Source File:

Education/scad/counting_array_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Case & 4-Color Array Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Sarah has $50, buys 2 shirts at $15 each. Ch** Standard: CCSS.MATH.3.OA.D.8	$20	[1][5] [ENTER] [2] [×] [5][0] [x⇄y] [-
2	**A baker bakes 40 loaves, sells 16, packs rem** Standard: CCSS.MATH.3.OA.D.8	6 loaves per bin	[4][0] [ENTER] [1][6] [-] [4] [÷]
3	**5 boxes with 8 pencils each, plus 12 extra l** Standard: CCSS.MATH.3.OA.D.8	52 pencils	[5] [ENTER] [8] [×] [1][2] [+]
4	**Three friends each buy a $4 snack, and split** Standard: CCSS.MATH.3.OA.D.8	$21	[3] [ENTER] [4] [×] [9] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Traditional calculators require (24 - 6) / 3; RPN evaluates naturally in sequence.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.